

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: MemC-Rev-Be.sil etc.

Creation date: 23-Sep-79 16:12:37

For: Spruce

26 total sheets = 25 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

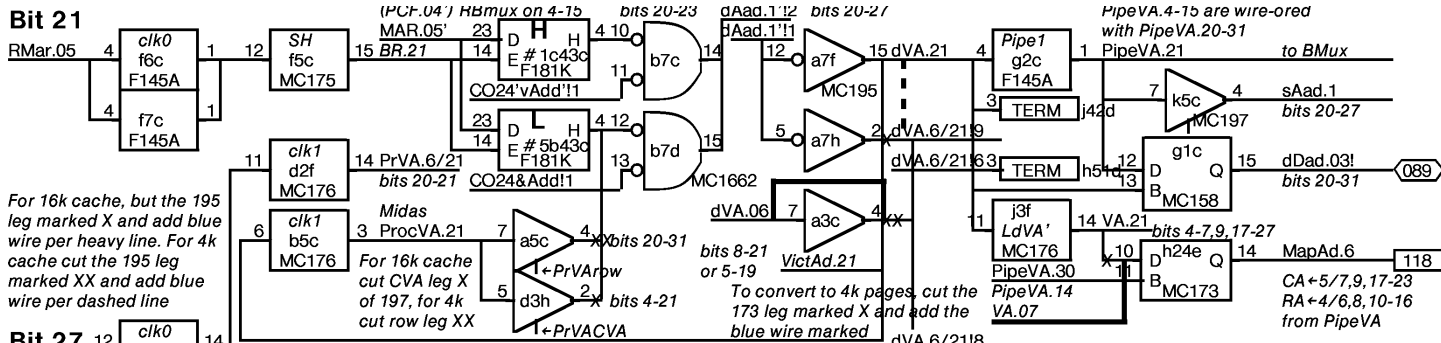
DORADO SCHEMATICS

Memory Control

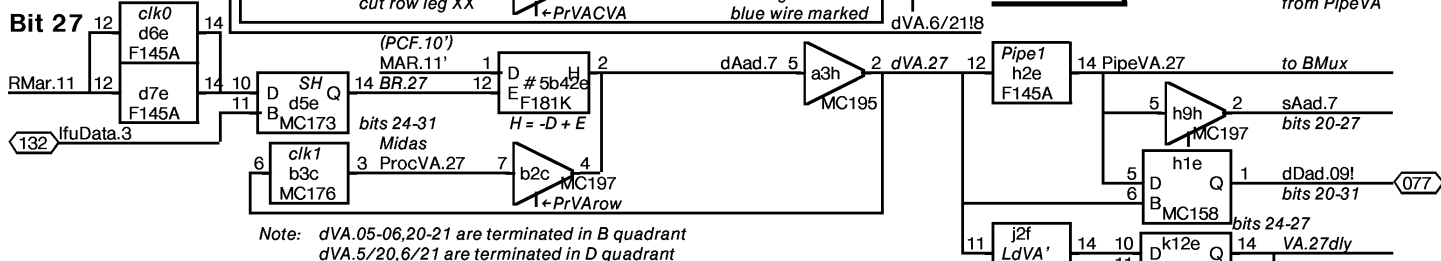
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Bit 21

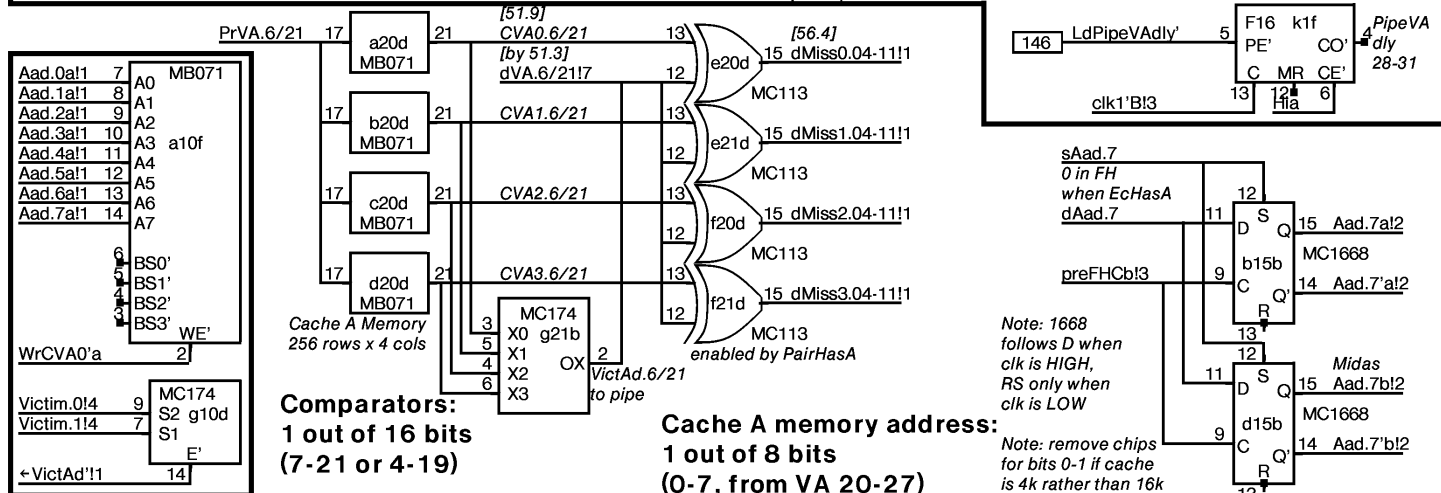
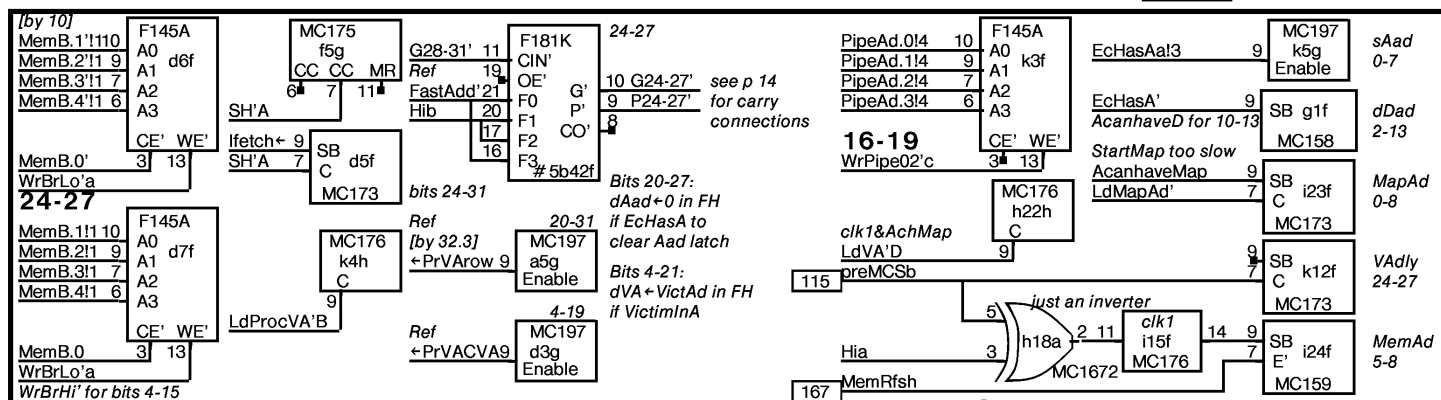
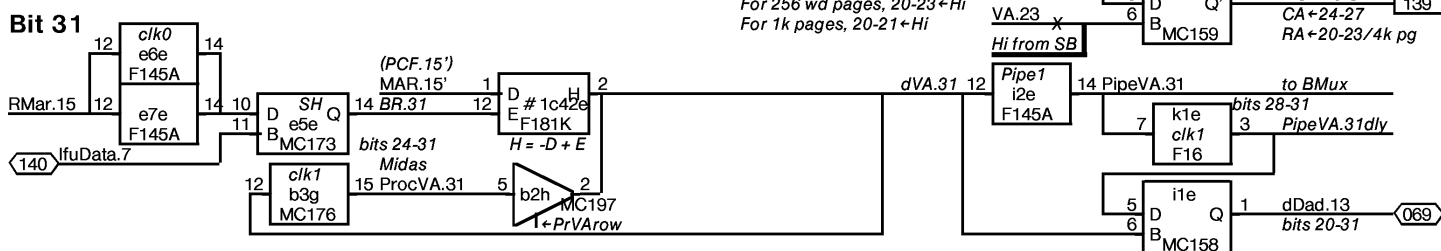


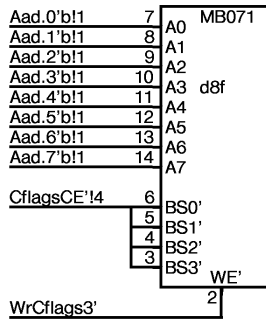
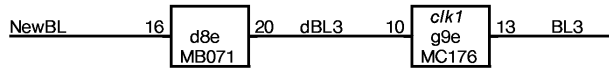
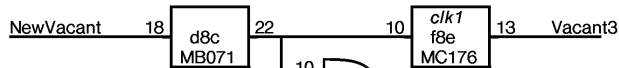
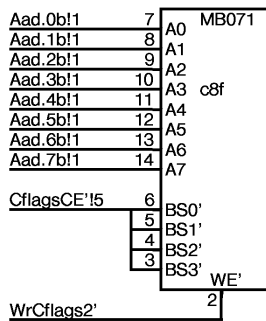
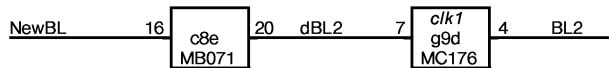
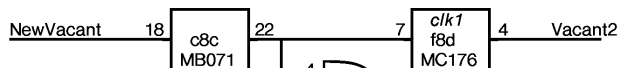
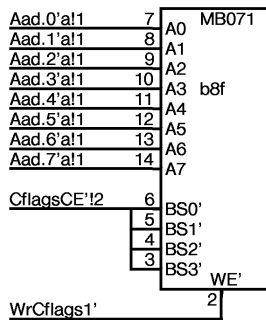
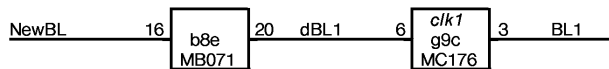
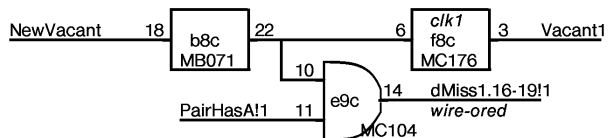
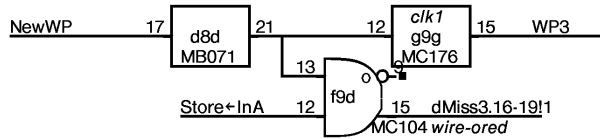
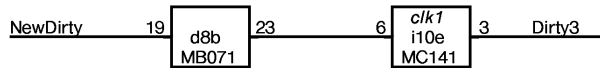
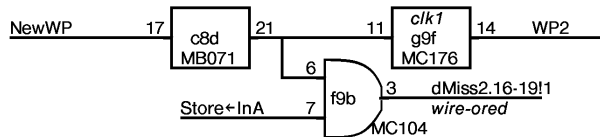
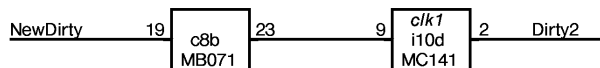
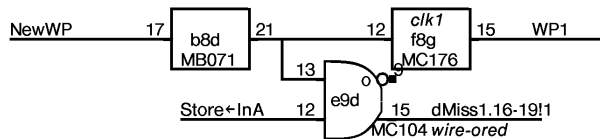
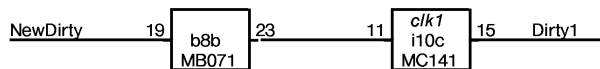
Bit 27

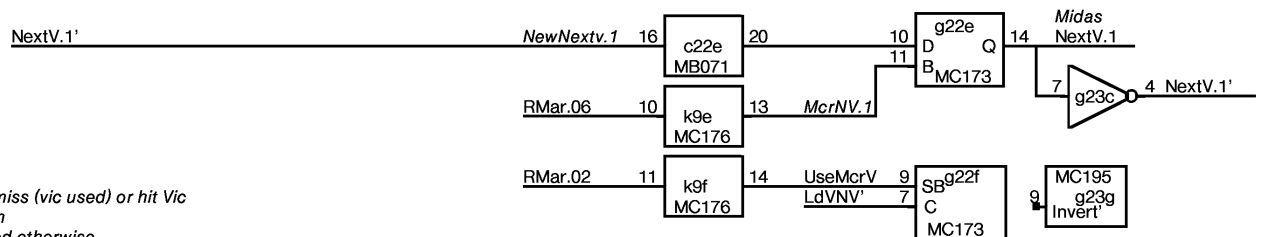
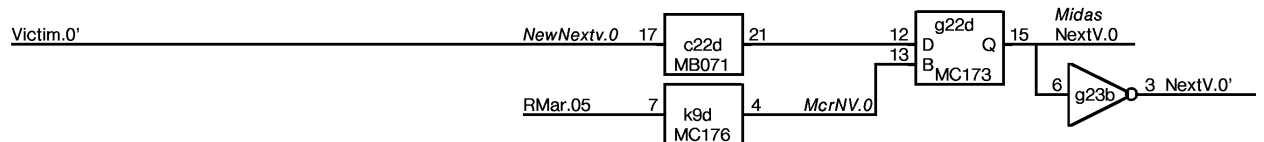
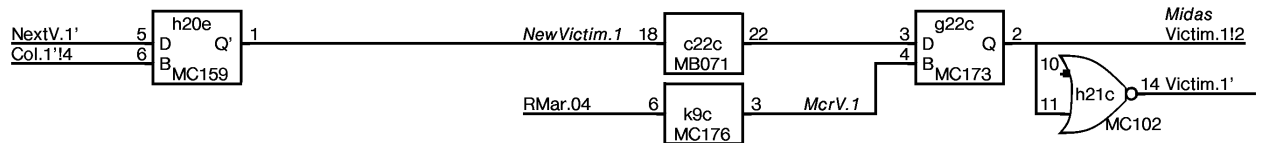
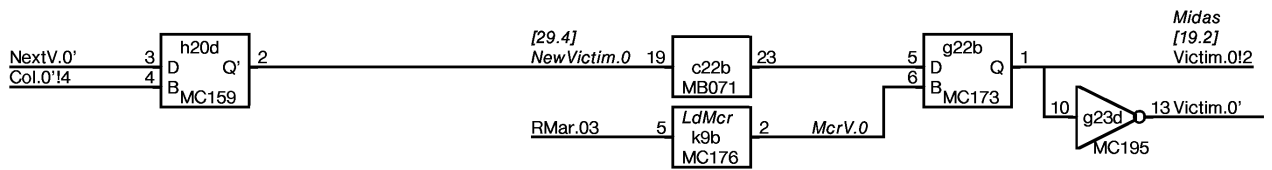


Main data path: 3 out of 28 bits (4-31)

Bit 31

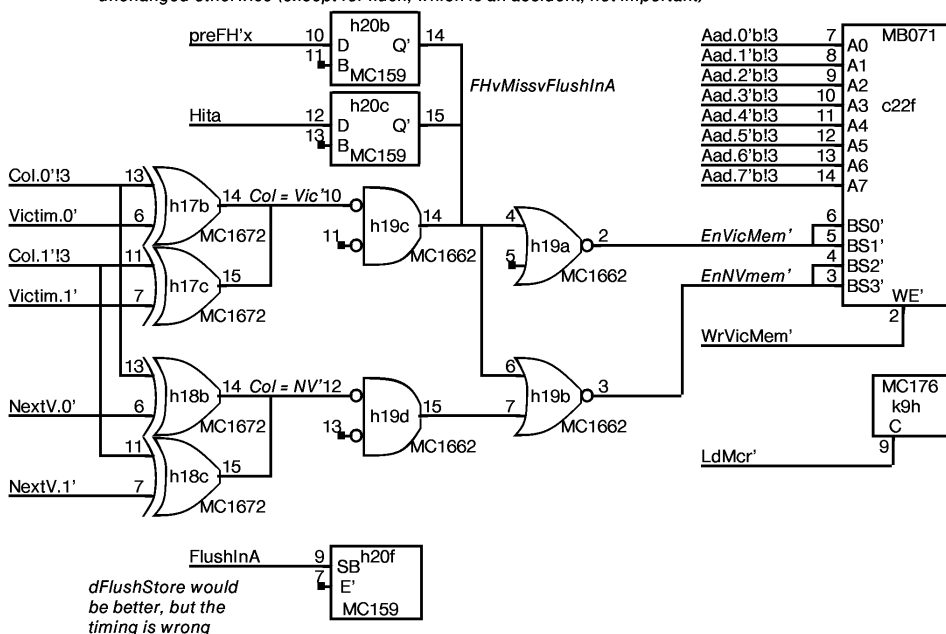






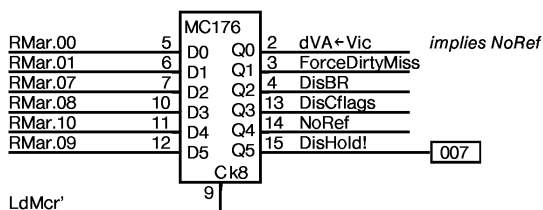
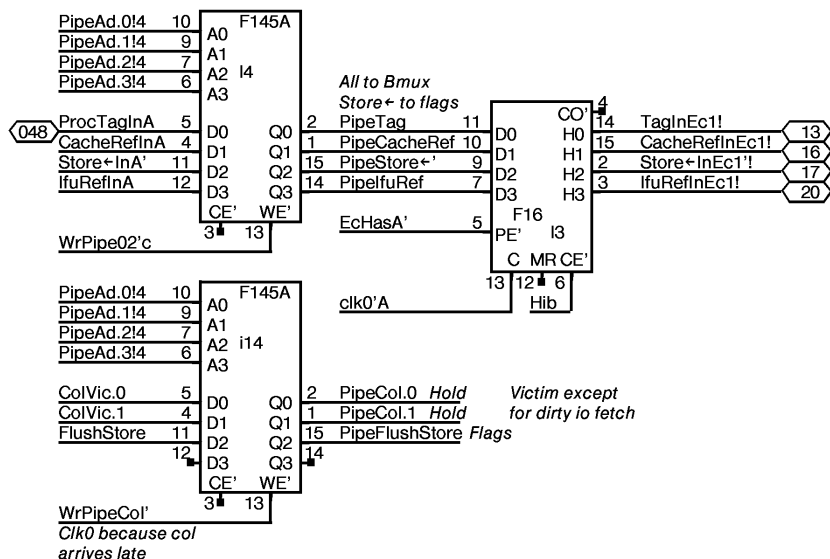
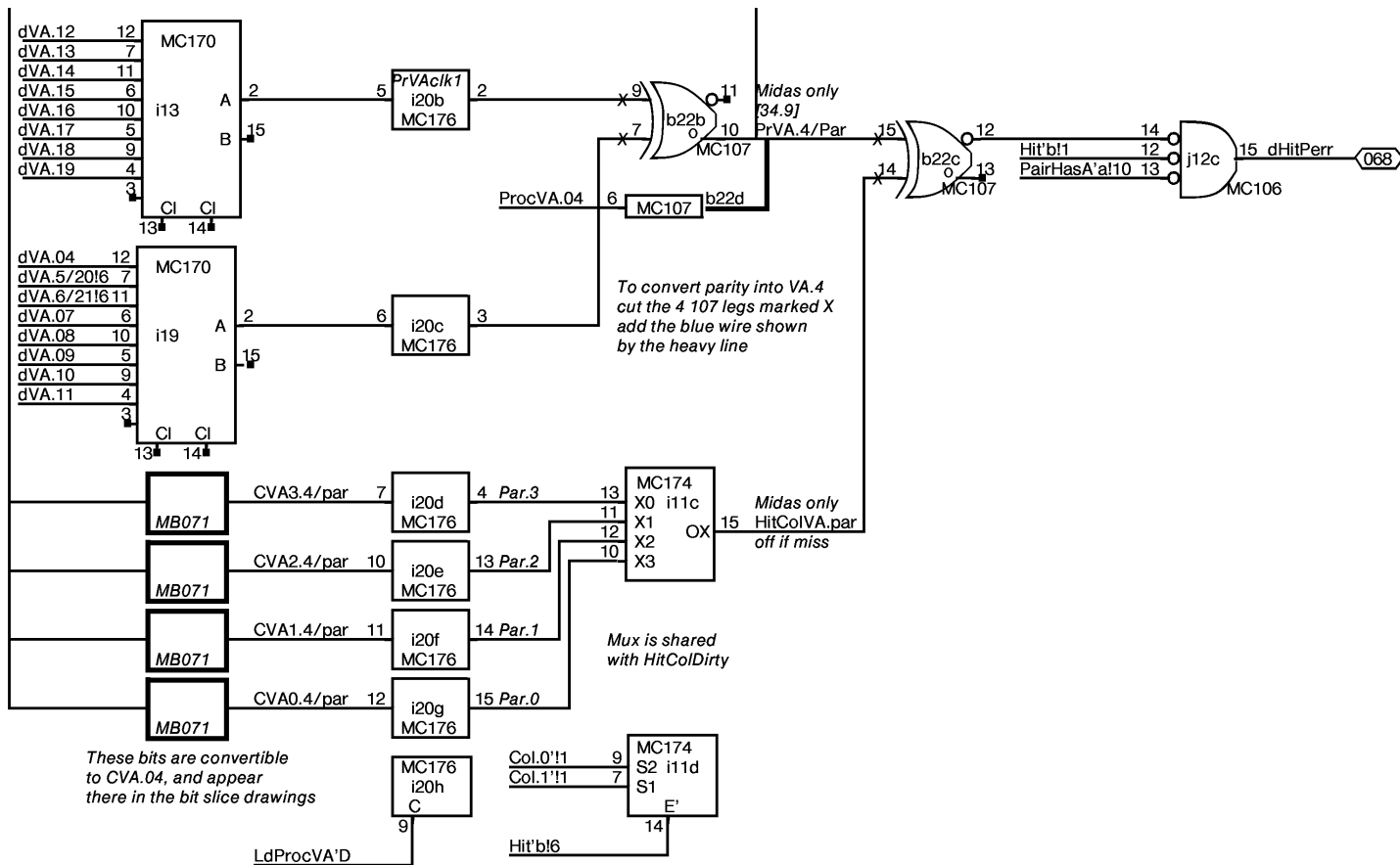
New Vic ← old NV if miss (vic used) or hit Vic
Col if flush
unchanged otherwise

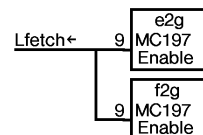
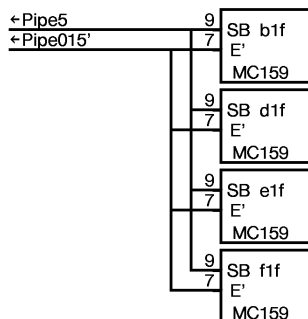
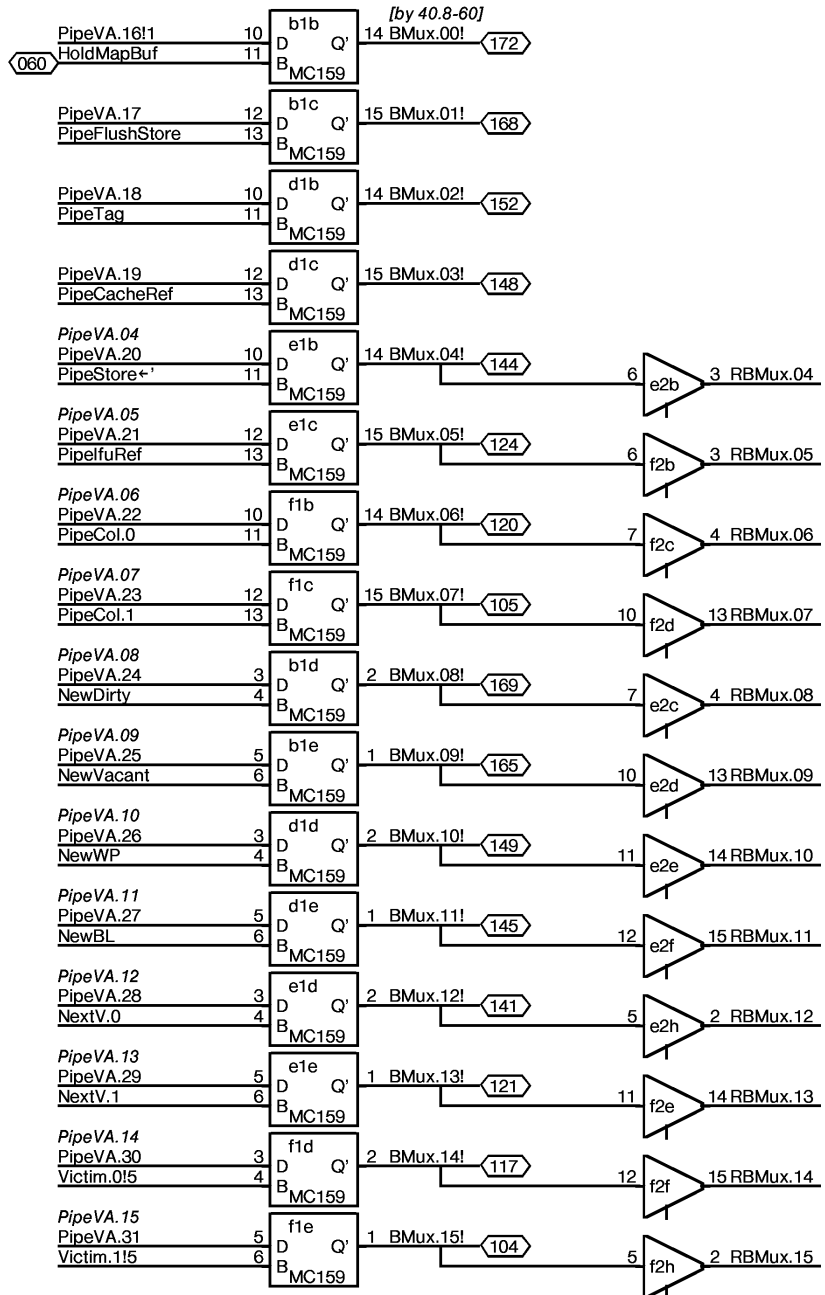
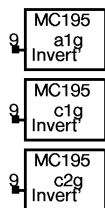
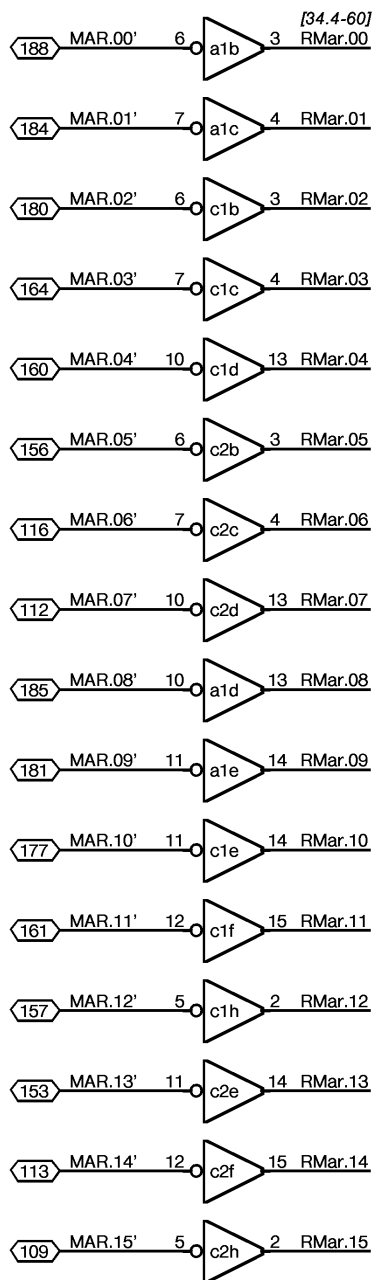
New NV ← something not = old Vic or old NV if miss or hit Vic (NV used) or hit NV
unchanged otherwise (except for flush, which is an accident, not important)

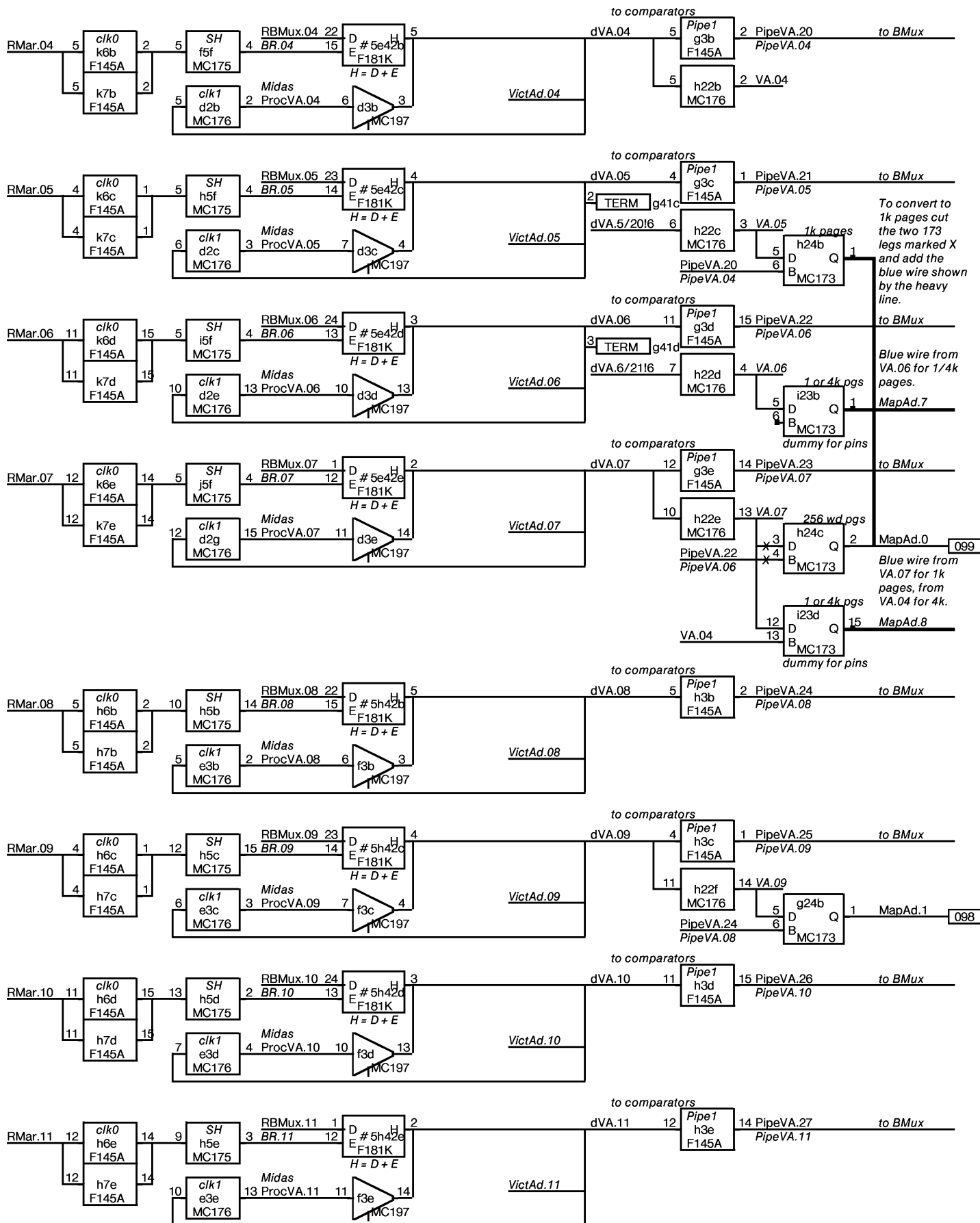


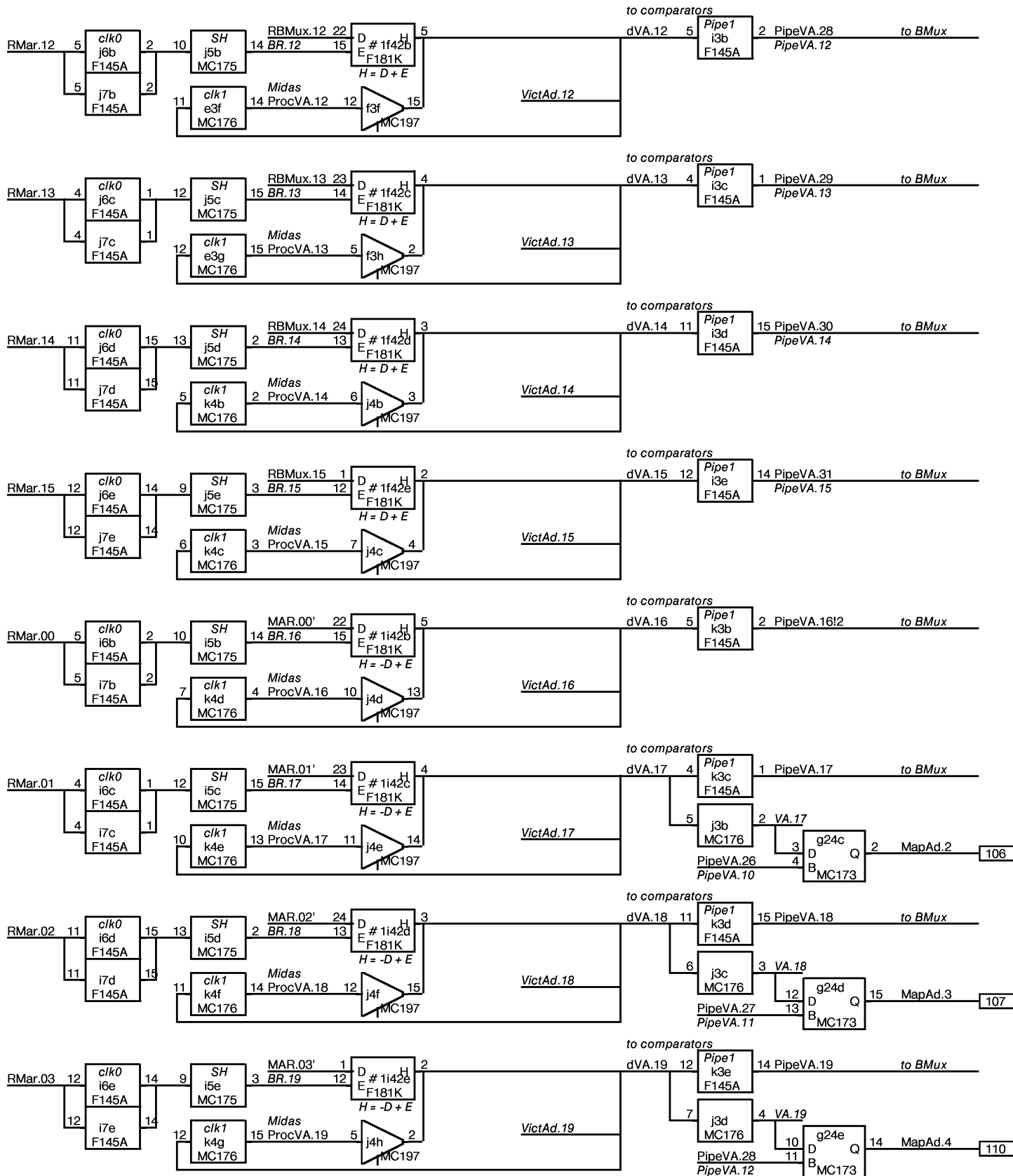
The Victim memory is written on a CacheRef, PreFetch or IfuRef that misses, and on any FlushStore or Flush+.

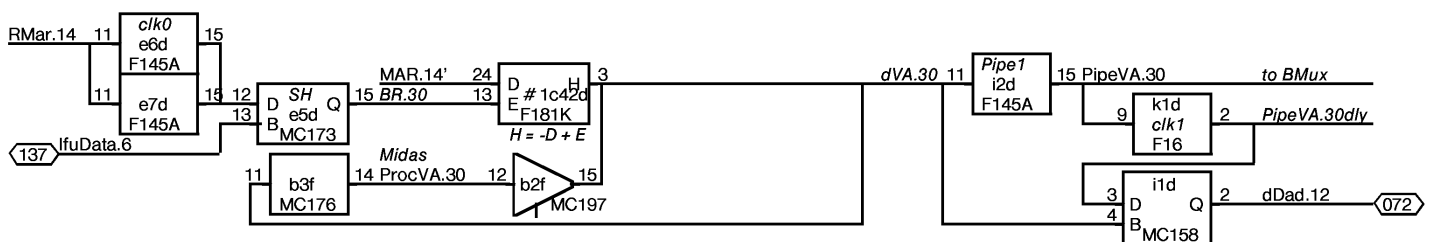
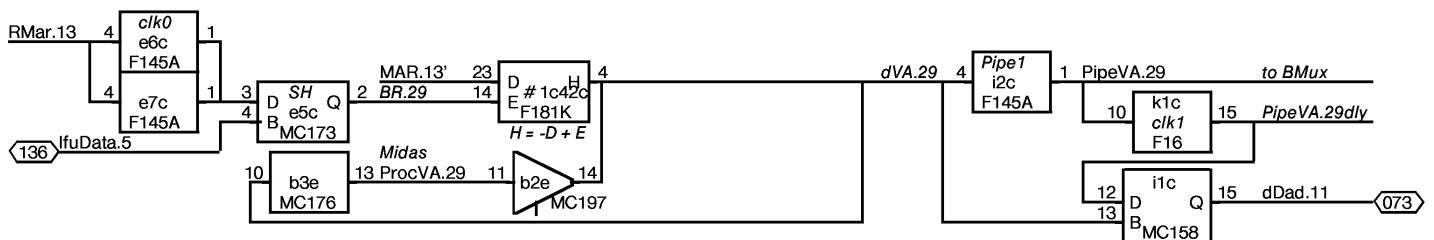
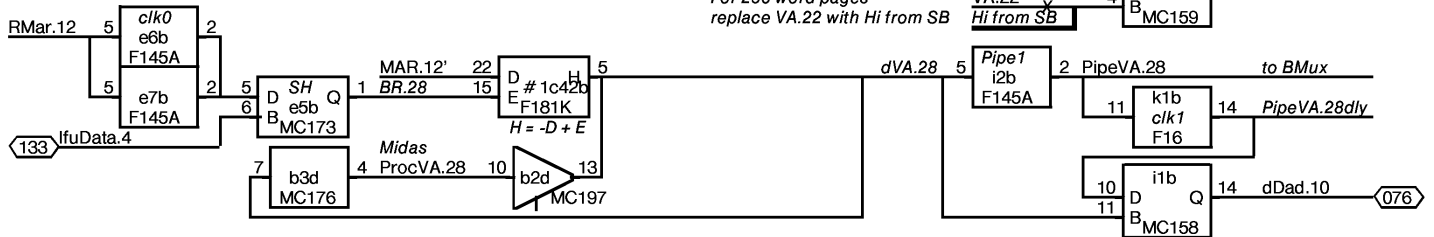
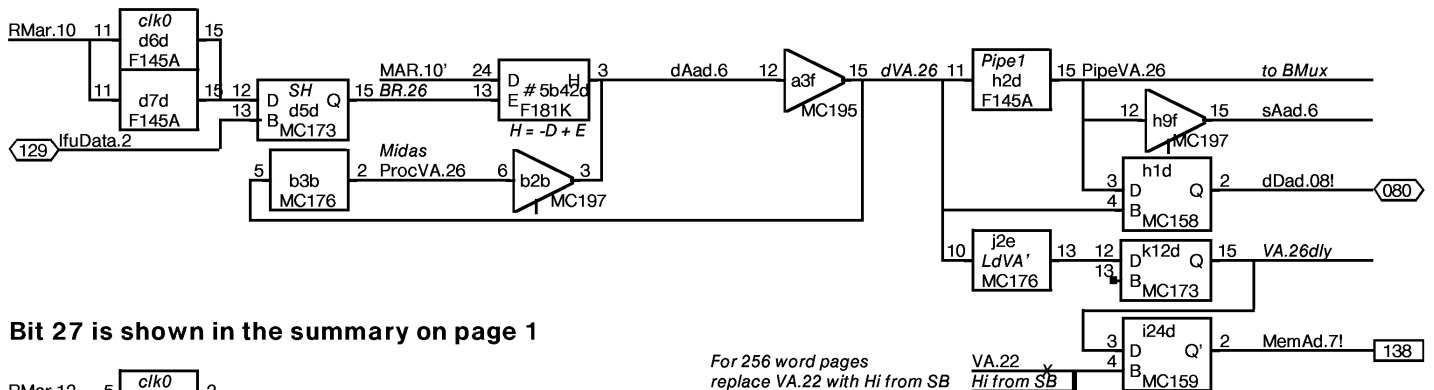
For a Flush+, Victim is first written with 0 on a miss or with the column of the hit. NextV is garbaged at the same time. On a dirty hit, a FlushStore follows, smashing Victim and NextV again.



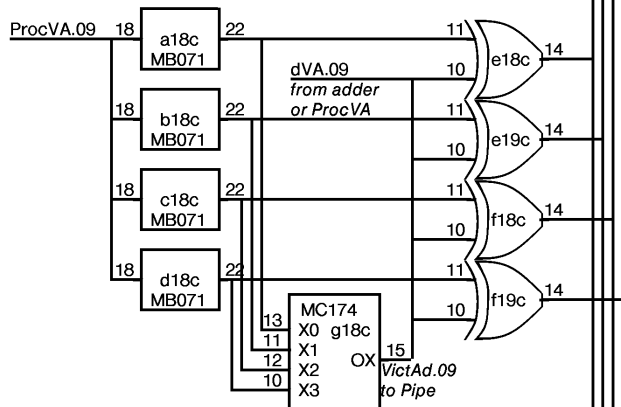
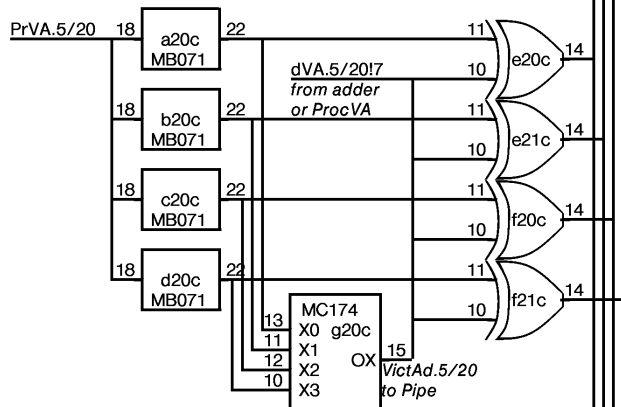
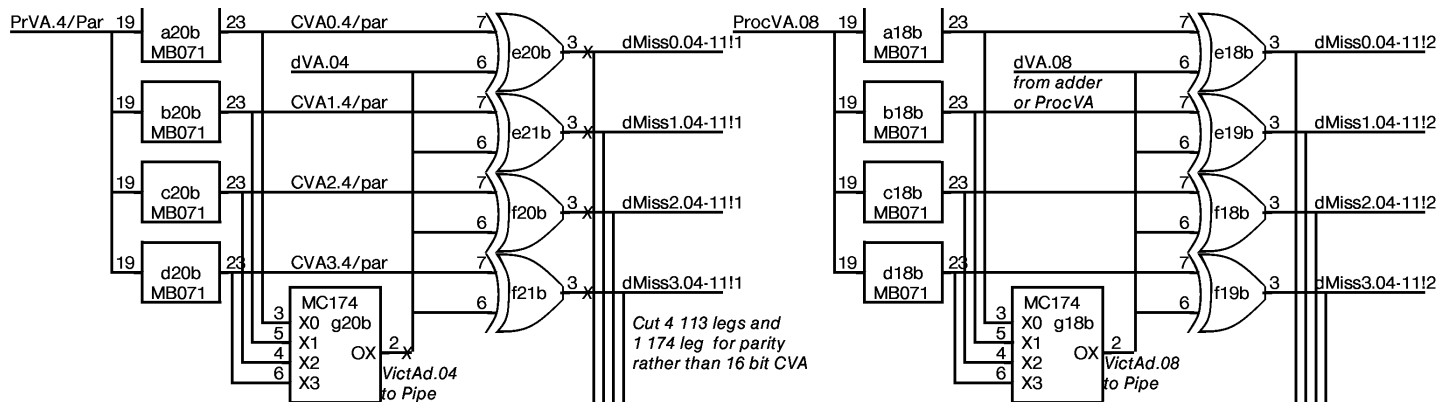




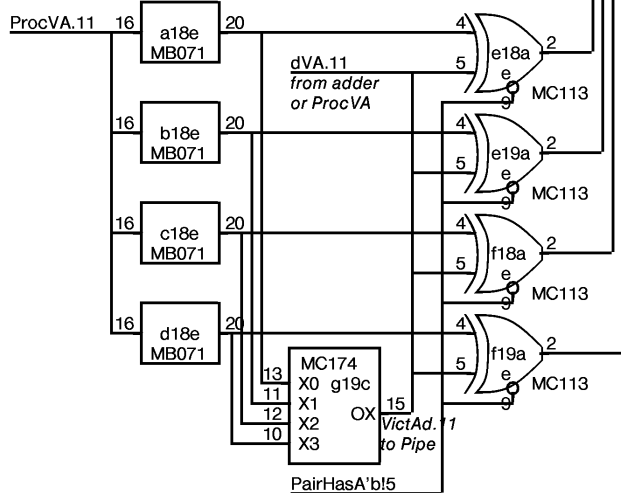
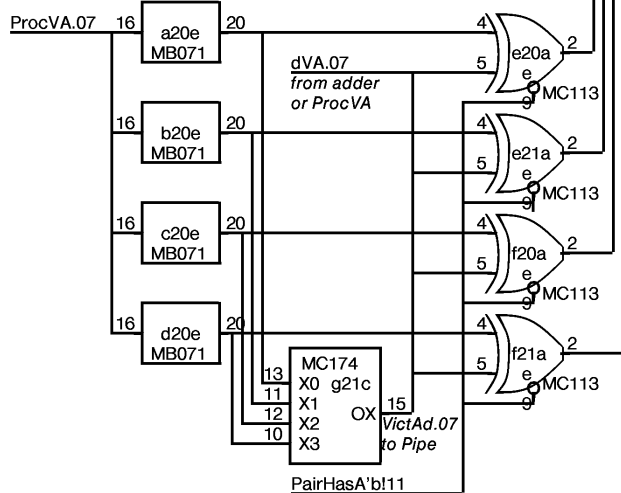
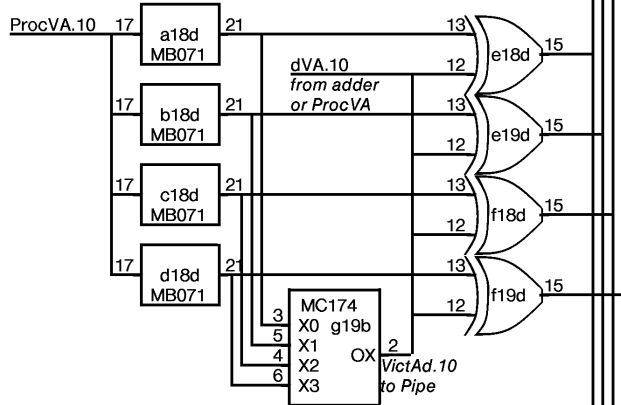


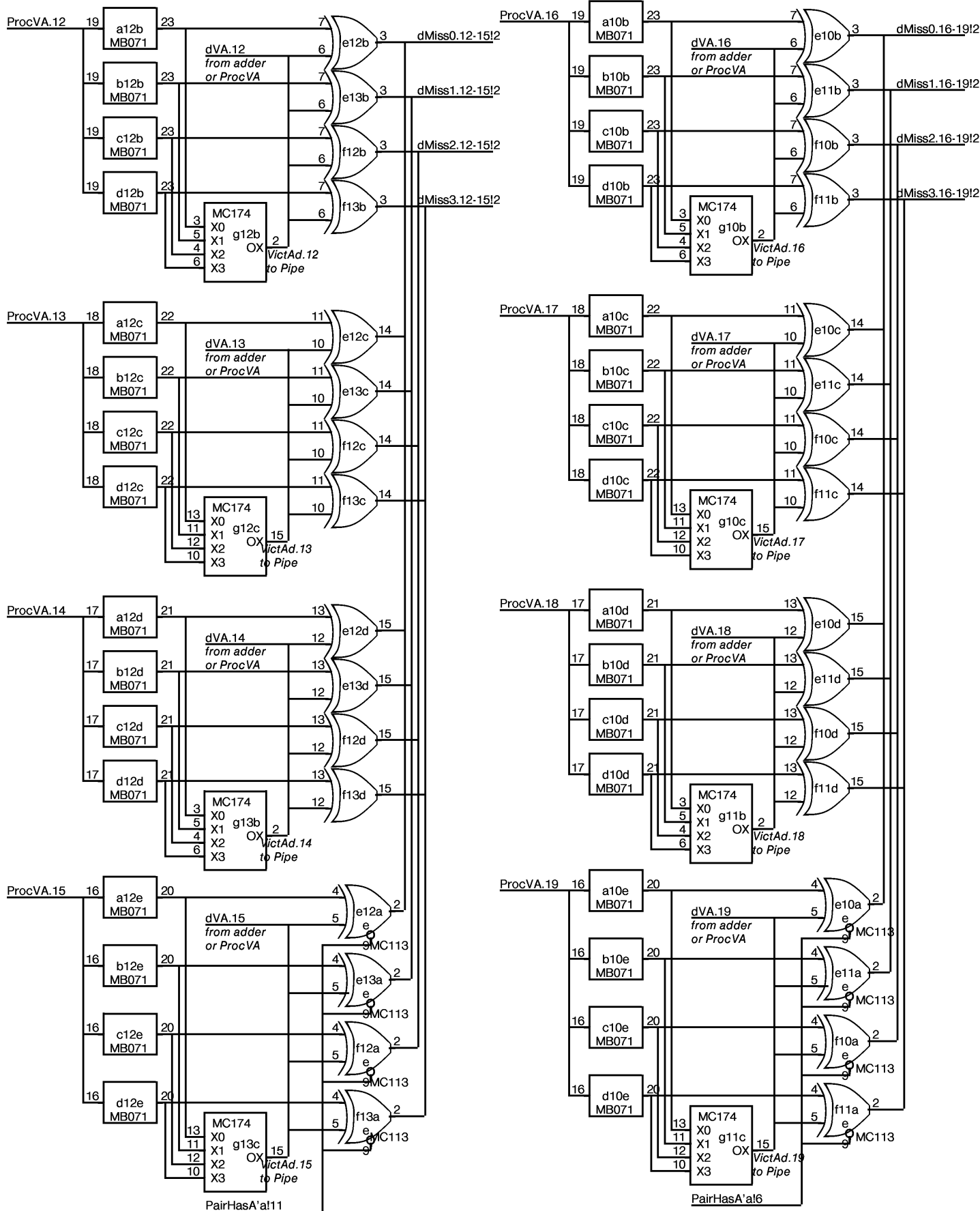


Bit 31 is shown in the summary on page 1

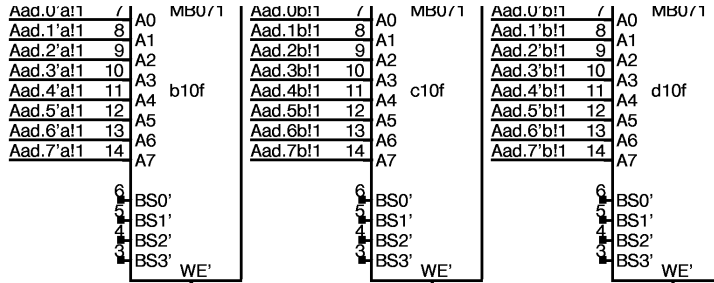


Bit 6 appears in the summary on page 1

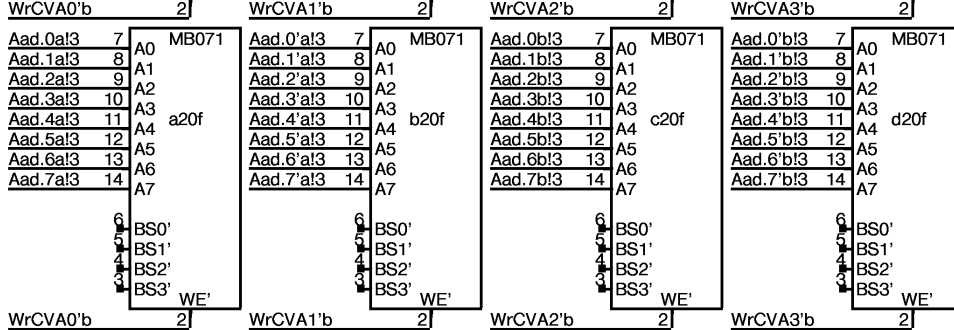
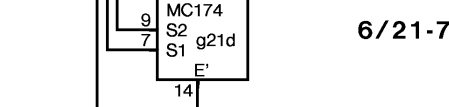
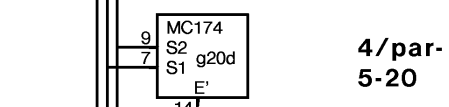
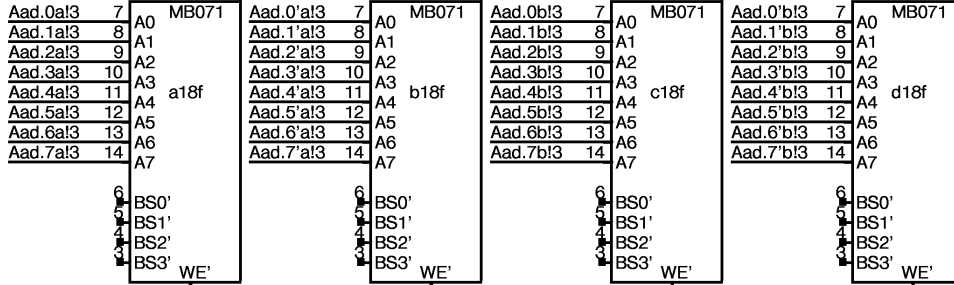
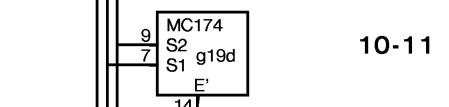
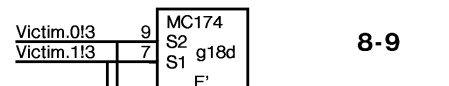
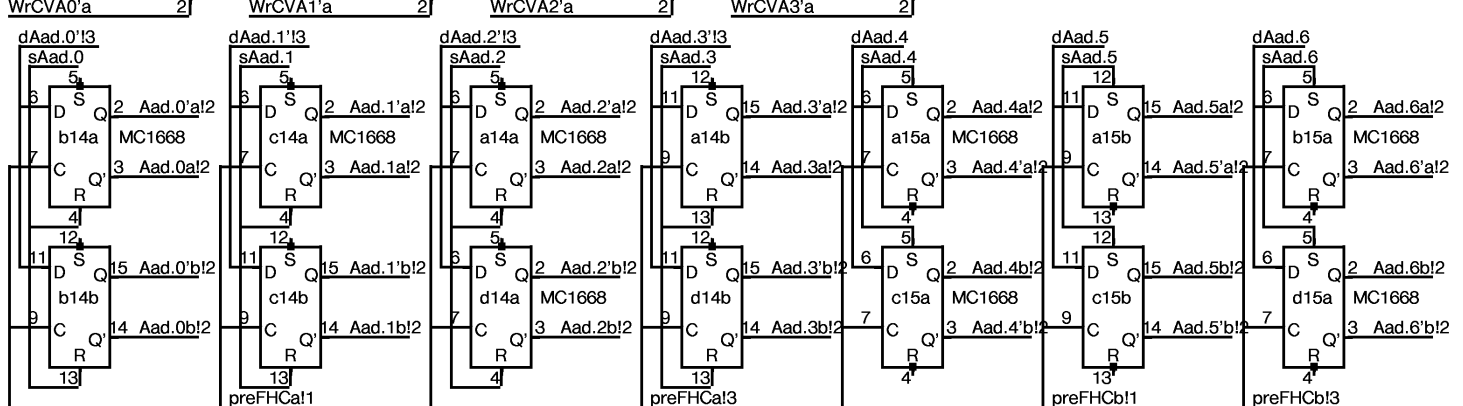
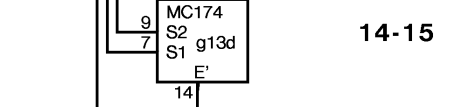
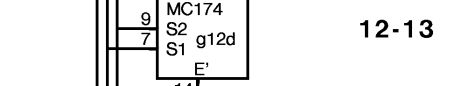
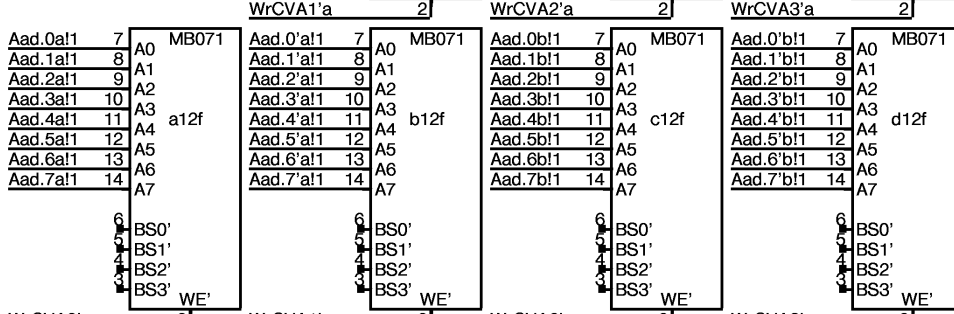
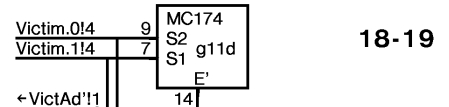


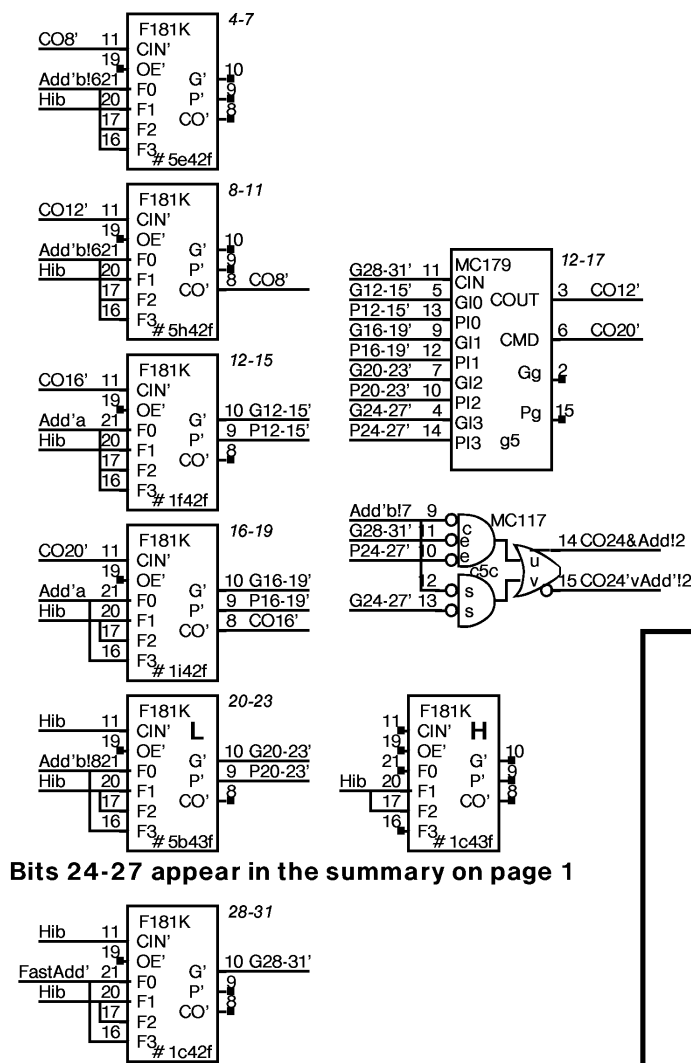


This chip's
addressing
appears in
the summary
on page 1



THIS VICTAD mux control
appears in the summary
on page 1 16-17





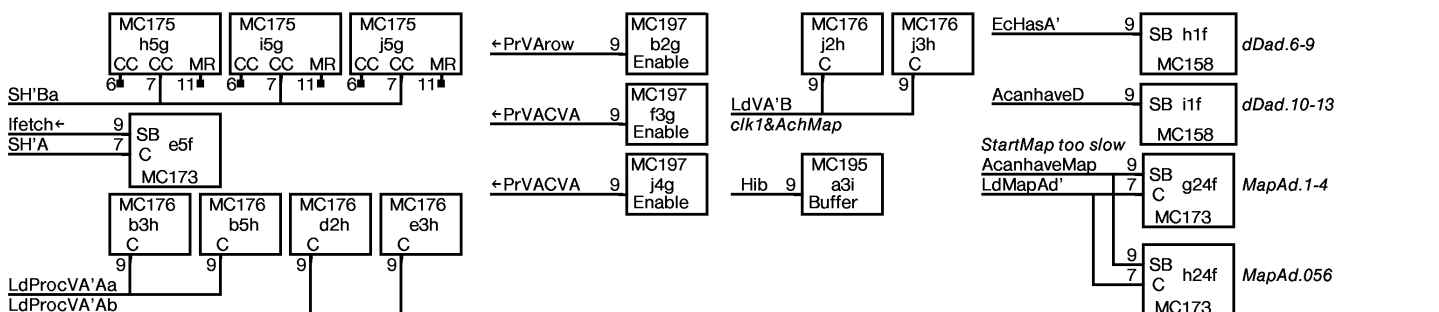
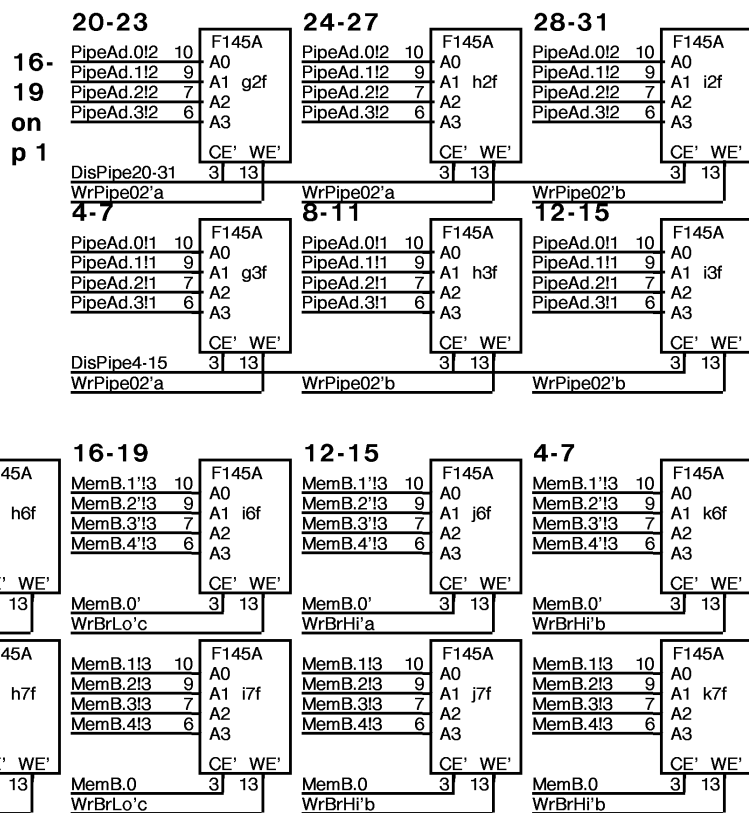
Timing: 28-31 6.5
24-27 4.1 + 4.7 = 8.8
20-23 4.1 + 3.8 + 1.9 = 9.8
16-19 4.1 + 6.0 + 4.7 = 14.8
12-15 4.1 + 6.0 + 3.6 + 4.7 = 18.4
8-11 4.1 + 6.0 + 4.7 = 14.8
4-7 4.1 + 6.0 + 3.6 + 4.7 = 18.4

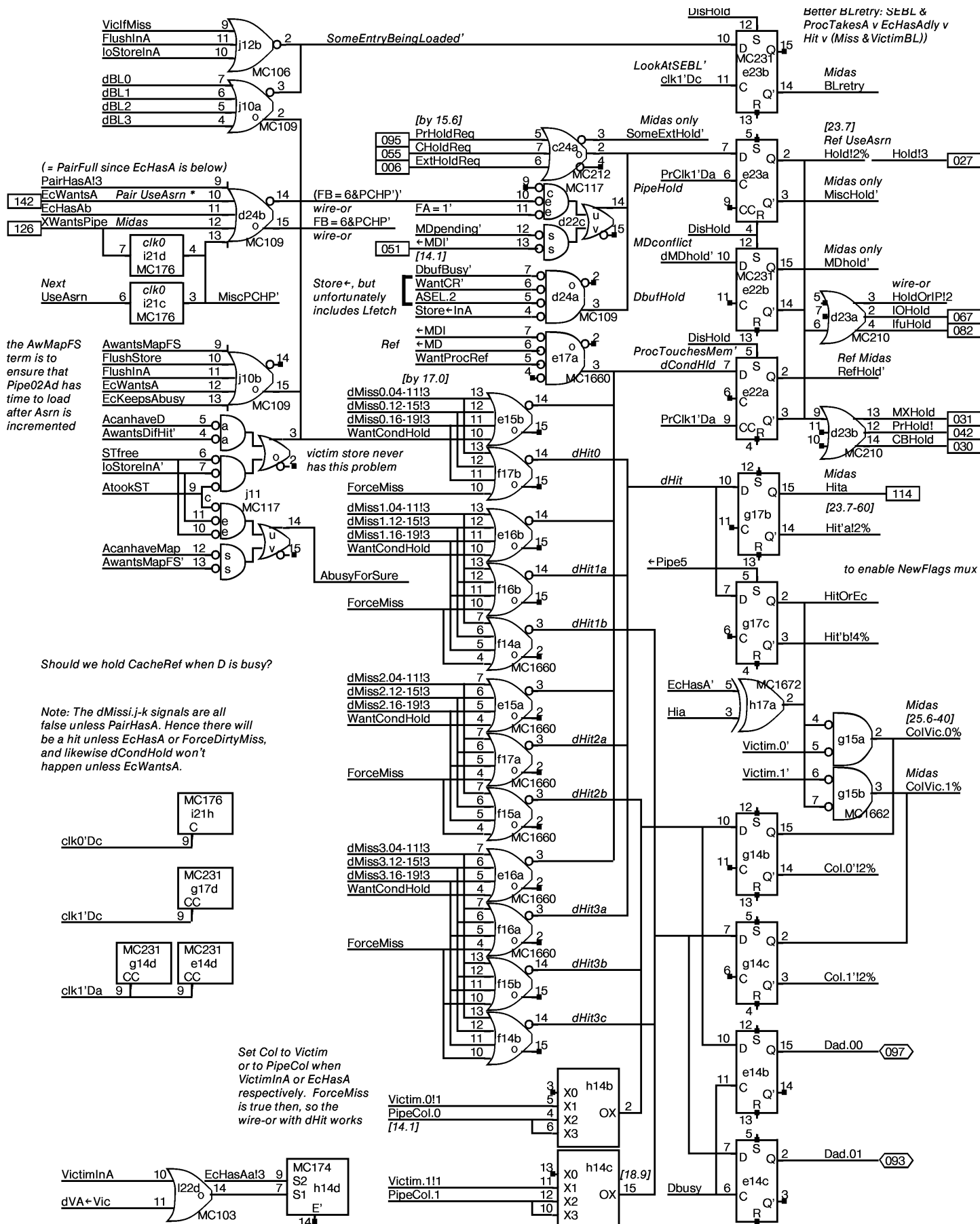
dAad. A RAM outputs are 2.8 + 10 = 12.8 later or at 22.6

dVA, for comparators, must meet the RAM outputs

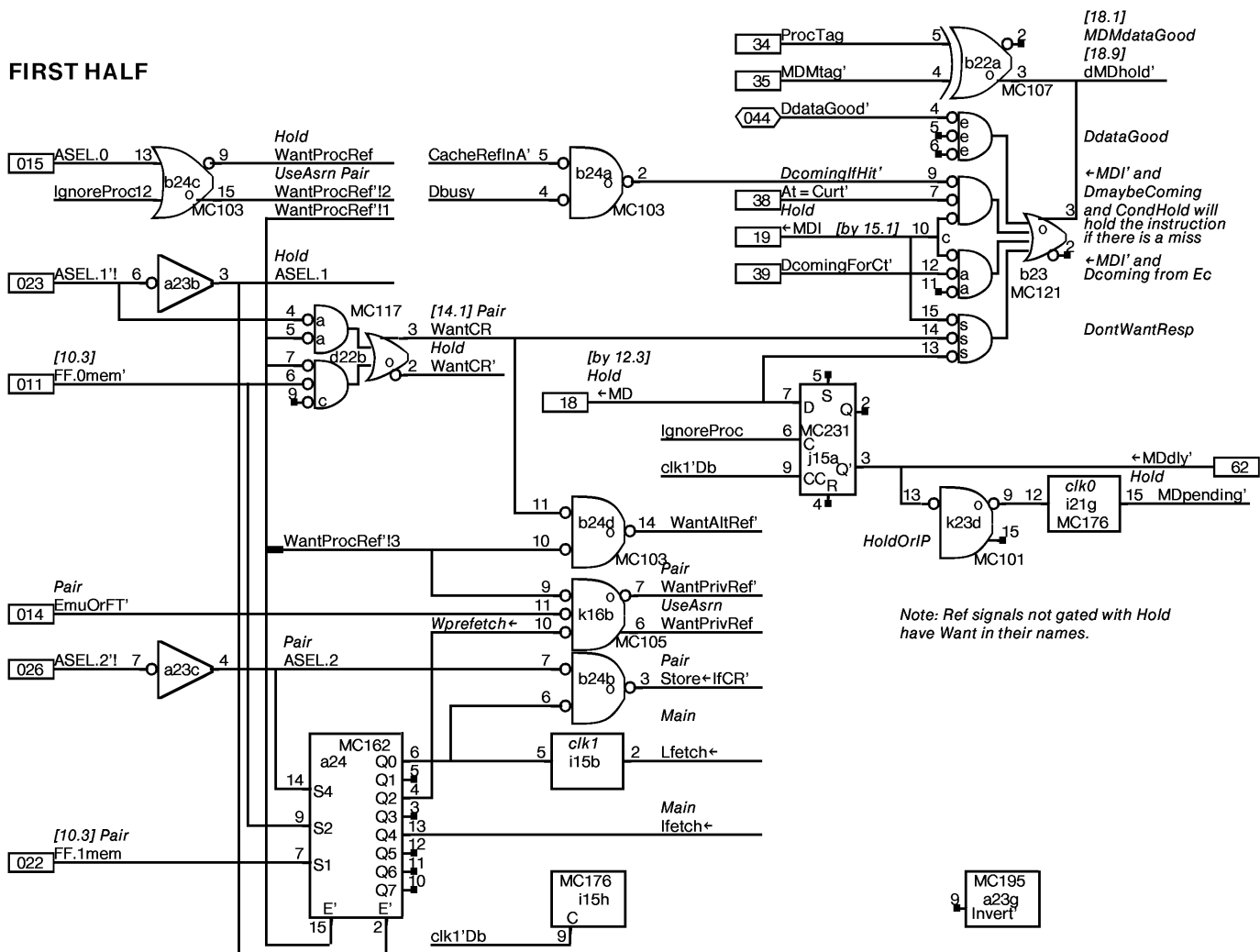
Control: D = RBMux/Mar' Add' = > 0 from 181K, or F = 1111
E = BR Add = > D + E, or F = 0100 on 4-15
-D + E, or F = 0110 on 16-31

Bits 24-27 appear in the summary on page 1

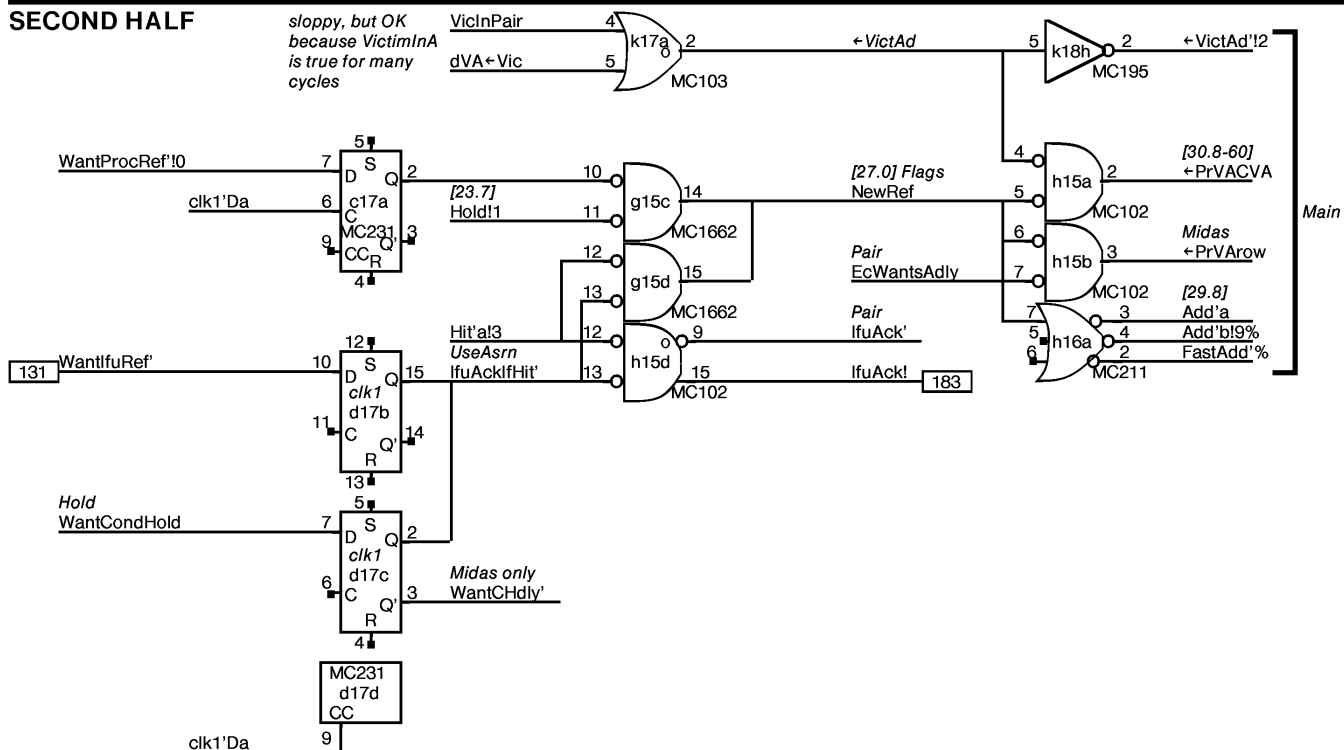


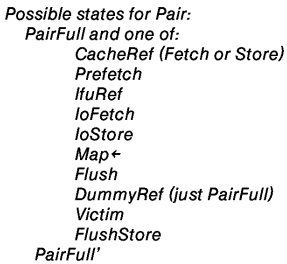


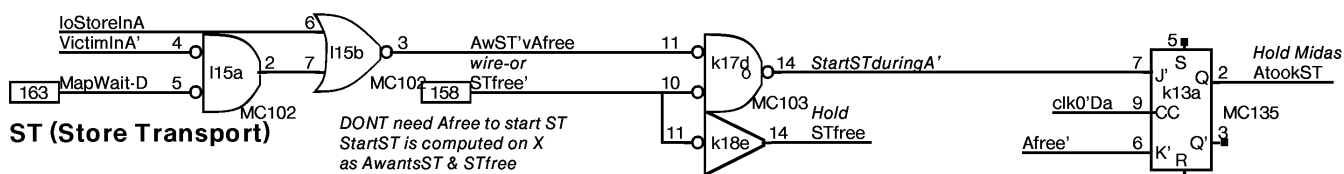
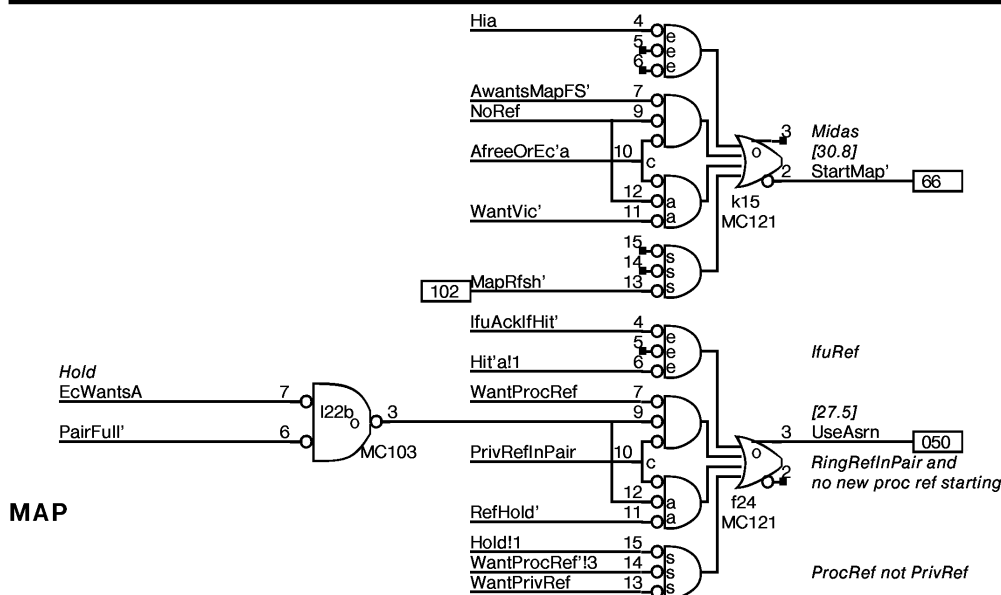
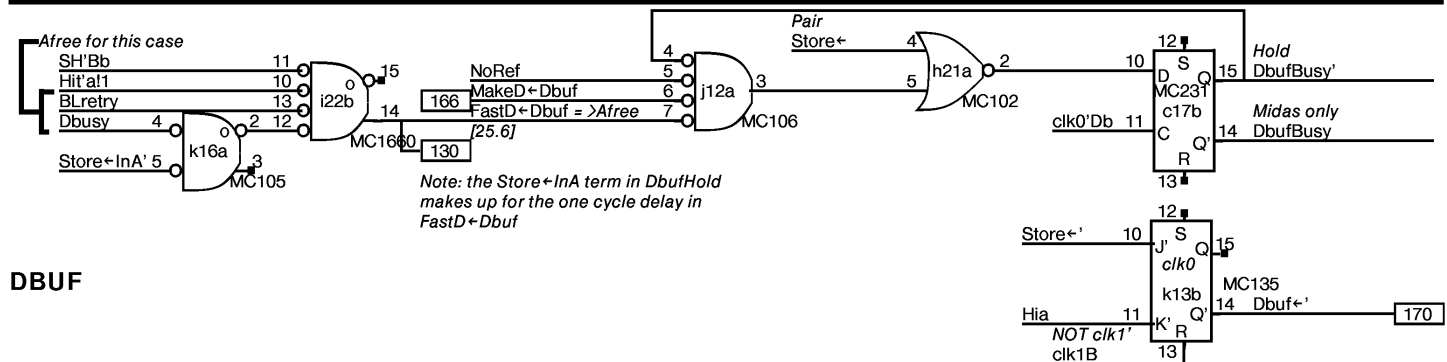
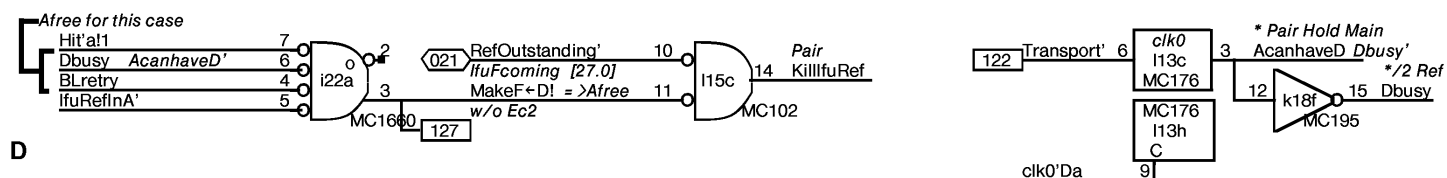
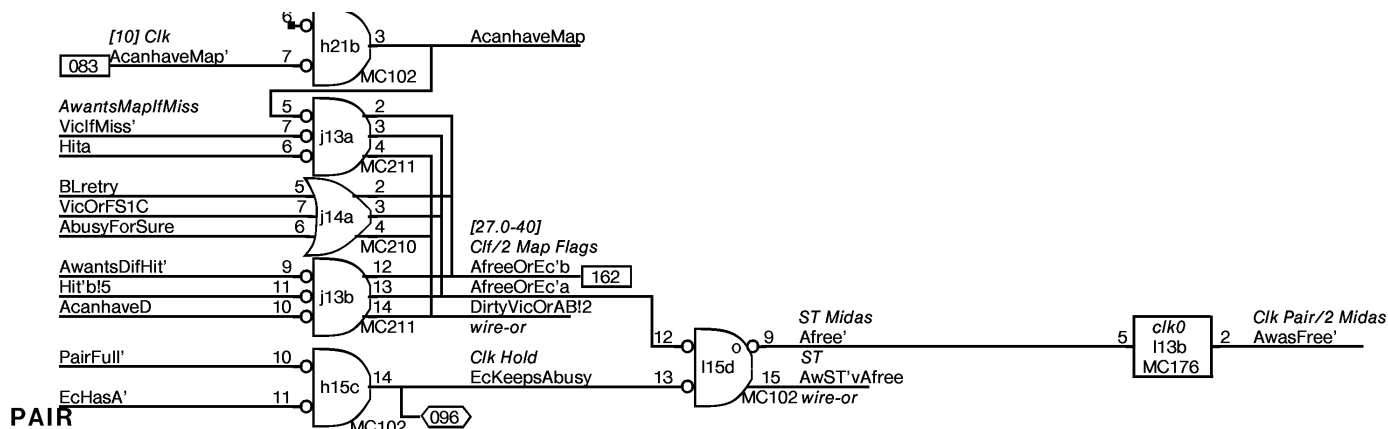
FIRST HALF

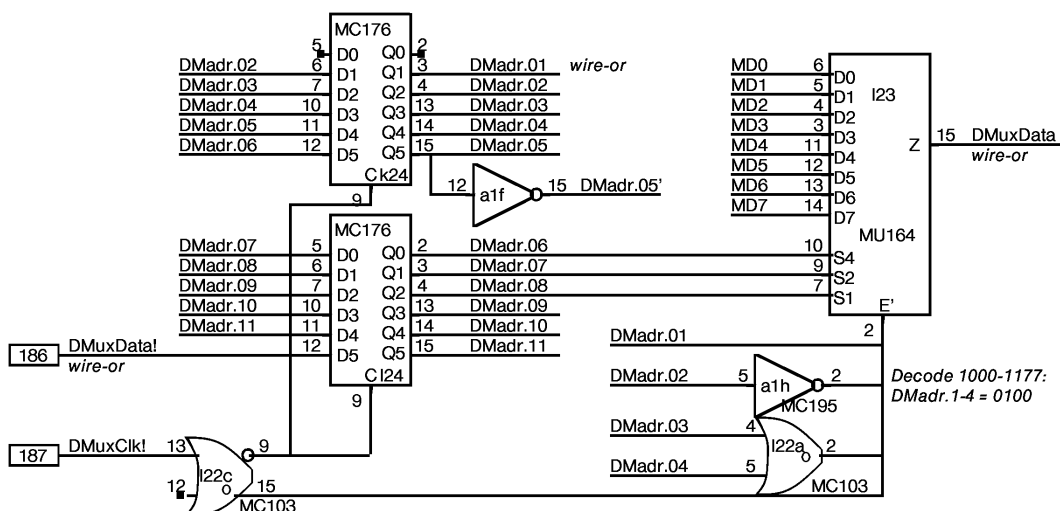
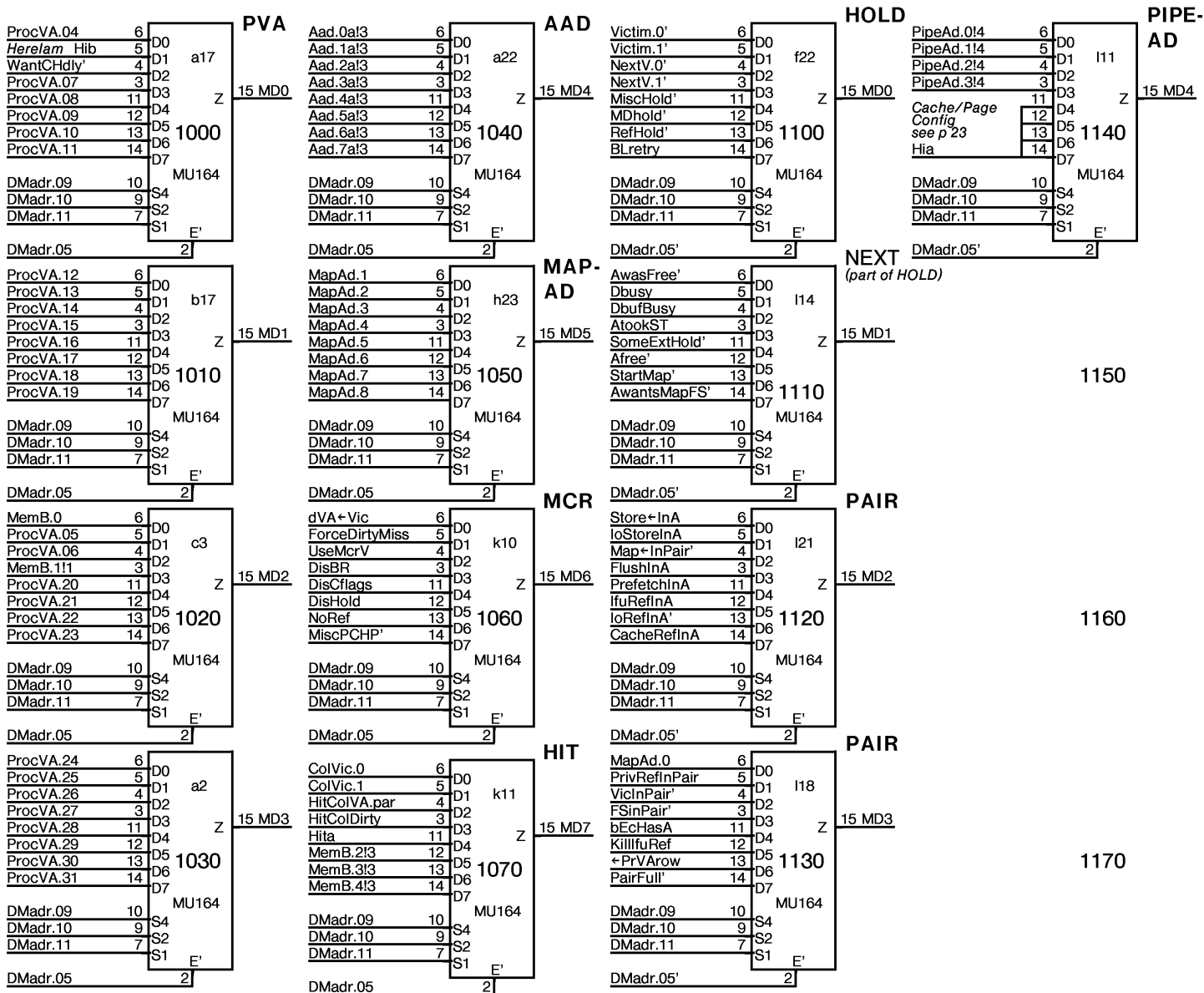


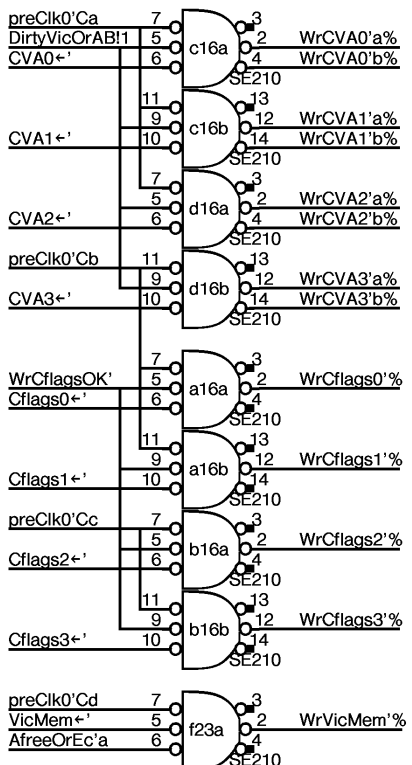
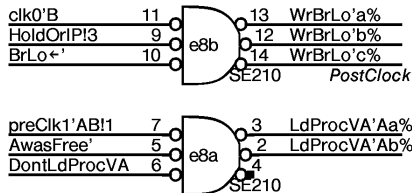
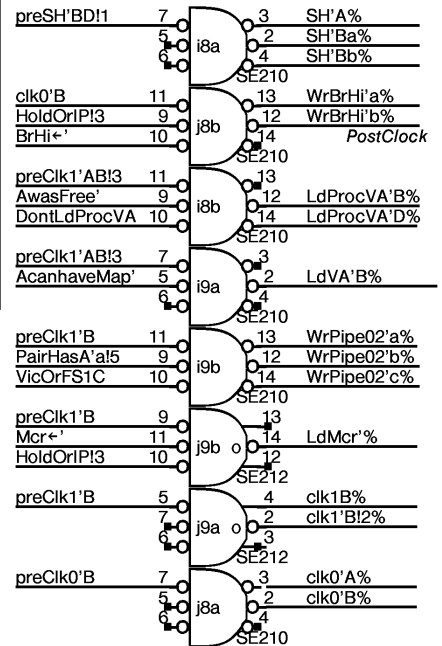
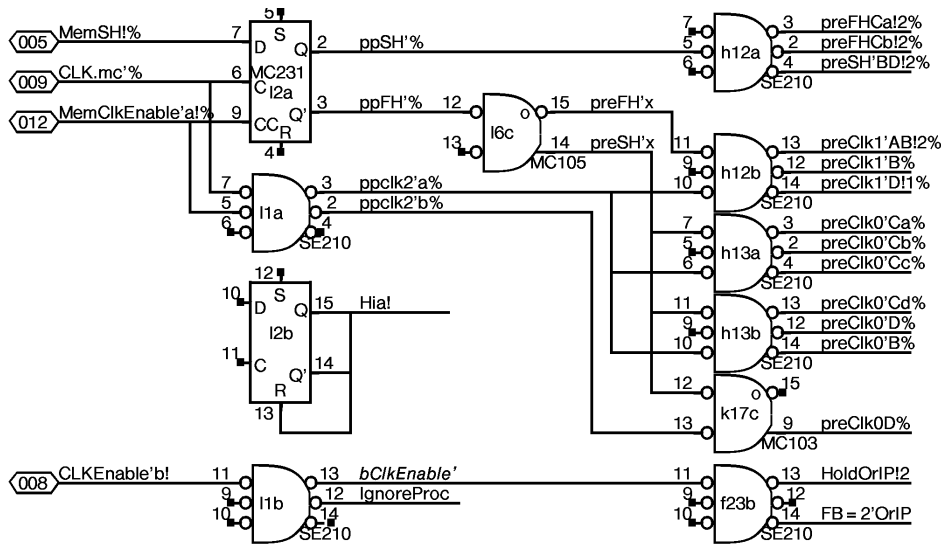
SECOND HALF



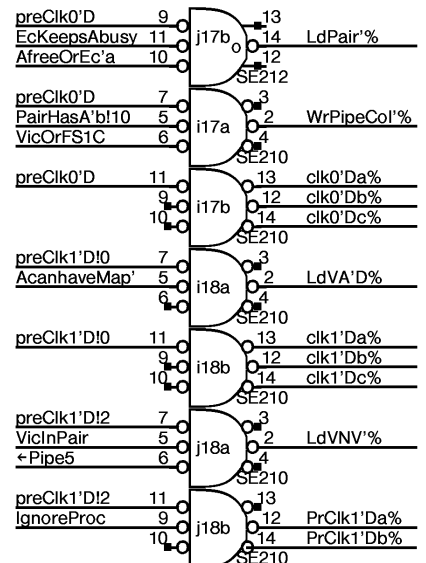
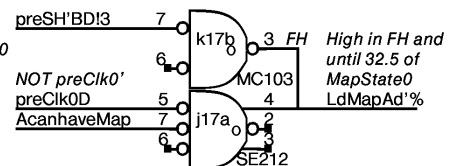








MapAd must be
low by 12.8 before clk0 of StartMap
to: let row MapAd through by 6 before clk0
high from clk0 or StartMap to 27.5 after
to: hold row MapAd
low by 2.8 before clk0 after StartMap
to: let col MapAd through by 4 after clk0



XXXX Mar 0/8		BMux 0/8		18 16		IfuData		Mar 7/15		BMux 93		dDad.02-10		CLKEn CLK.mem 20		CikSync RunRfsh	
A	a	b	c	d	e	f	g	h	i	j	k	l	B				
1	RMar 0189 DMadr.2,5 195i	BD 0189	RMar 2-4,10-12 195i	BMux driver			2-5 158	6-9 dDad 2-13 158	10-13 158	MemB+37 +Pipe015' 1662	PipeVAdly 28-31 F16	Clock 210	1				
2	MU ProcVA 24-31	Proc 197	5-7,13-15 195i	176	RBMux 4,8,12 197 5-7,13,15 197			Pipe 20-31 PipeAdl2		VA Hlib 141	PipeAdl3 231	ovh 231	2				
3	dVA buf 22-27 195	VA/dr 26-31 176	MU PrVA56,20-23 MemB.0-1	ProcVA & dr			4-13 197	5/20,6/21	197	Pipe 4-15 PipeAdl1	17-27 176	Pipe 16-19 PipeAdl4	PipeTag F16	3			
4	24-27 181k	Adder 181k	28-31 181k	4-7 181k	Adder 181k	12-15 181k	8-11 181k	Adder 181k	16-19 181k	ProcVA & dr 197 141 19 176		145	4				
5	ProcVA & dr 197 20 25 176		Cout.24' 117	IfuDR/BR 24-31 173		BR latch 4,20-23 175	Cout12,20' 179	5,8-11 175	BR latch 6,16-19	7,12-15 175	sAad.0-5 197	disPipe/3 +Pipe15' 103	5				
6	20-23 181k	Adder 181k	20-23 181k	Base register RAMs			Base register RAMs			Base register RAMs			MemB.0! Dec.pSHx 105	6			
7	dVAbuf.20-23 dVA.56/2021 195i	dAad.0-3 mux 1662		24-27 145	28-31 145	20-23 145	MemB.1-4 101	8-11 145	16-19	12-15	4-7 145	FF PRclk1 176	7				
8	Aada	Aad'a	Aadb	Aad'b	LdProcVA' WrBrHl' 210	Cflags reg 176	Cf+RMar 100	New 174	Clocks		Mcr 176	decod- 162	8				
9	Cache flags			Miss-Cflags 104			Cflags reg BL + 176	sAad 6-7 197			ing PRclk1 176	9					
10	A bits 16-19			Comparators			Victad 105	Cflags 105	Dirty 141	SEBL WantCH 109	Mcr	161	10				
11				16-19			16-19 174	HitColDirty Parity 174	AbusyFS WantCH 117	MemB.2-4 +5	MU PipeAd +4	MU	11				
12	A bits 12-15			Comparators			Victad 174	Pre 210	DirtyVicOrAB 174	dDbufBusy! dHitPerr SEBL 106	VAdly 173	+	12				
13				12-15			12-15 210	clocks 210	dVA.12-19 parity 170	Afree 211	Dbuf+ST AtookST 135	Next/2 MB.0 176.0	13				
14	2/3	0	1	2/3	Dad.0-1 231	Col 231	Col.0-1 231	PipeCol 174	145	210	DirtyloF!	MU 104	14				
15	4/5 1668	6/7	4/5	6/7 1668	Hold 1660		ColVic.0-1 NewR,HorEc 1662	IfuAck! +PrVA/2 EcKeepsAB 102	Lfetch EcwAdly ABdly MCS 176.1	+MdDly' 231	StartMap 121	Afree! AwST/2 KillIfuRef 102	15				
16	210	CVA/Cflags clocks			210	1660	Hit 171	211	102	121	WantPrivRf! FastD+Dbuf 105	+	16				
17	MU PrVA 4,7-19 +2	WantPRdly DbufBusy 231	IfuAcklfHit 231	ProcTchMem 1660	1660		Hit 231	HitOrEc' Col = Vic' 1672	Clocks		+preCik0 MapAdLd StST,+VicA 103	+	17				
18	A bits 8-11			Comparators			Victad 1672	Col = NV' preMCS' New 1672			Pair/3 Next/2 +VictAd 195i	MU Pair	18				
19				8-11			8-11 1662	VNV EnVNV 1662	dVA.4-11 parity 170	WantVic! VicM+ Rdlna AwDifHit 102	ForceMiss VicfMiss 109	VicOrFS1C (EcHasA) 100	19				
20	A bits 7, 20-21, parity or bits 4-7			Comparators			Victad 159	stuff 159	Parity PrVackl1 176	101	PairInA (EcHasA) 101	101	20				
21				4,7,20-21 or 4-7			4,7,20-21 or 4-7 102	Vic.1' IoFiA AchMap dDbufBusy 102	BPCHP,Pr/2 JsaAsrndly MdPend. 176.0	Pair		MU Pair	21				
22	MU Aad	BMdH,dPerr dVA.4/par 107	VNV	WantCR,CR dMDc,PipH 117	RefH, MdH 231	MU xHold/3 BLretry VNV	VNV 173	VA 4-7,9 176	FastD+Dbuf MakeF+D! 1660	St+ ,IfuRef Abusy 105	Pref,loRef DntLdPrVA! 104	Midas/2! D+Dbuf Pcol+Vic 103	22				
23	ASEL12 mkD+CD 195i	dMDhold 121	RAM	xxHold dMiscH 210	MiscH, BLrty 231	WrVicMem IP buf 210	NextV' Vic.0' FF-5-7' 195i	MU 78 173	78 173	dVic/FS 117	Pair/3,MDd (HoldOriP) 101	Midas 164	23				
24	WantRef decodes 162	WantPR! DcomingIH St+icr,wAR 103	xxHold CacheRef' 212	dDbufHold PCHP 109	FB decode 171	UseAsrn 121	1-4 173	MapAd 173	MemAd 159	Ref decodes 161	overhead 176		24				
C	a	b	c	d	e	f	g	h	i	j	k	l	D				
Exxx	+MDI +MD Asel		FF.0-3		PRhold FF.4-7		MemBase		MemAd 5-8								
XEROX PARC		Project Dorado		Reference Layout		File MemC22.sil		Designer Lampson		Rev Be		Date 7/01/79		Page 22			

CACHE CONFIGURATIONS

4k CacheConfig = 3			16k CacheConfig = 2			16k without parity CacheConfig = 1		
Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)
a03	195	3x 4x cut 5-6 from 5/20-6/21	a03	195	3-6 4-7 connect 5-6 to 5/20-6/21	a03	195	3-6 4-7 connect 5-6 to 5/20-6/21
a07	195	2-15 3-4 connect 20-21 to 5/20-6/21	a07	195	2x 4x cut 20-21 from 5/20-6/21	a07	195	2x 4x cut 20-21 from 5/20-6/21
a05	197	3x 4x read ProcVA.20-21 for CVA	d03	197	2x 15x read ProcVA.20-21 for row	d03	197	2x 15x read ProcVA.20-21 for row
b14	1668	remove disconnect Aad.0						
c14	1668	remove disconnect Aad.1						
e20	113	3x	e20	113	3x	b22	107	7x 9x 14x 15x 6-10 connect 4 to 4/par disconnect Perr reporting
e21	113	3x	e21	113	3x			
f20	113	3x	f20	113	3x			
f21	113	3x keep parity from comparators	f21	113	3x			
g20	174	2x keep parity from CVA	g20	174	2x			
			l11	164	12x make CacheConfig = 2	l11	164	11x make CacheConfig = 1

PAGE SIZE CONFIGURATIONS

256 words PageConfig = 3			1k words PageConfig = 2			4k words PageConfig = 1		
Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)	Position	Chip	Cut (X) or wire (pin-pin)
i24	159	4x 6x 11x 13x 4-6-11-13-9 MemRA+0 0 0 0	i24	159	11x 13x 11-13-9 MemRA+0 0 22 23			MemRA+20 21 22 23
			h24	173	3x 4x 1-2 MapAd.0+4-5 for 6-7	h24	173	10x 12x 10-3 12-5 MapAd.5,6+5,7 for 20,21
			i23	173	3x 10x 3-5 10-12 MapAd.7,8+6,7 for 22,23	i23	173	3x 10x 3-5 10-13 MapAd.7,8+6,4 for 22,23
			l11	164	14x make PageConfig = 2	l11	164	13x make PageConfig = 1

Note: Muffler signals 1144-1147 specify the configuration as follows:

1144-1145 are CacheConfig

1146-1147 are PageConfig

and the meaning of their values is as listed in the tables above

Need information here about the two missing holes

CURRENTLY NONE